MATH 202 Your name: _____

Learning target I1, version 1

Compute approximations of definite integrals using numerical methods. Decide when each method gives overestimates vs. underestimates.

Consider the function $f(x) = \dfrac{\sin(x)}{x}$, which does not have an elementary antiderivative. Define the integral function $F(t) = \displaystyle\int_1^t f(x)\,dx$.

(a) What does the Second FTC say about the relationship between F and f?

(b) Use appropriate technology, such as Geogebra or a Riemann sum applet, to approximate $F(2)$ using left, right, midpoint, and trapezoid sums with 10 subintervals:

- $L_{10} =$
- $R_{10} =$
- $M_{10} =$
- $T_{10} =$

Report your answers with at least 6 decimal places of accuracy.

(c) Which of these approximations are overestimates and which are underestimates? How can you tell?

(d) Use your calculations from part (b) to approximate $F(2)$ using Simpson's rule.